

## VALENTIA OBSERVATORY, Ireland

WMO: 039530

Lat: 51.9383N

Lon: 10.2410W

Elev: 30

StdP: 100.97

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -0.4       | 1.2 | -4.4                          | 2.6 | 1.9  | -2.8 | 3.0 | 3.1  | 16.5                  | 7.8  | 14.6 | 8.3  | 2.0                   | 60   | 0.625 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 4.6                    | 21.2            | 16.6 | 19.4 | 15.9 | 18.3 | 15.5 | 17.7                | 19.5 | 17.0 | 18.3 | 16.3 | 17.6 | 3.5                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.0                            | 12.2 | 18.2 | 16.3 | 11.7 | 17.6 | 15.7 | 11.2 | 17.0 | 49.9          | 19.5 | 47.7 | 18.3 | 45.8 | 17.7 | 22.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 13.3          | 11.6 | 10.2 | DB                         | -2.6 | 24.5               | 1.8 | 2.1                                                | -3.9 | 26.0       | -5.0 | 27.2       | -6.0 | 28.4       | -7.4 | 29.9 | (4) |
| (5)               |      |      | WB                         | -3.3 | 19.0               | 1.9 | 1.4                                                | -4.6 | 20.0       | -5.7 | 20.8       | -6.7 | 21.6       | -8.1 | 22.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.1   | 7.6   | 7.5   | 8.1   | 9.5   | 11.6  | 13.7  | 15.3  | 15.3  | 14.2  | 12.1  | 9.5   | 8.2  |
|                                                                                  | DBStd       | 3.60   | 2.52  | 2.33  | 2.36  | 2.08  | 2.04  | 1.83  | 1.57  | 1.52  | 1.81  | 2.18  | 2.43  | 2.70 |
|                                                                                  | HDD10.0     | 367    | 79    | 73    | 66    | 32    | 7     | 0     | 0     | 0     | 0     | 7     | 37    | 66   |
|                                                                                  | HDD18.3     | 2656   | 333   | 302   | 316   | 266   | 209   | 139   | 97    | 95    | 124   | 193   | 266   | 315  |
|                                                                                  | CDD10.0     | 757    | 4     | 3     | 8     | 17    | 56    | 112   | 163   | 164   | 127   | 72    | 22    | 9    |
|                                                                                  | CDD18.3     | 4      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 2     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 13     | 0     | 0     | 0     | 0     | 1     | 4     | 3     | 4     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.7    | 5.6   | 5.4   | 4.9   | 4.5   | 4.4   | 4.1   | 3.8   | 3.9   | 4.2   | 4.7   | 5.0   | 5.6  |
| Precipitation                                                                    | PrecAvg     | 1500   | 174   | 128   | 117   | 93    | 92    | 89    | 92    | 109   | 126   | 162   | 170   | 166  |
|                                                                                  | PrecMax     | 2177   | 320   | 292   | 260   | 198   | 208   | 184   | 256   | 273   | 290   | 320   | 360   | 339  |
|                                                                                  | PrecMin     | 990    | 26    | 2     | 19    | 24    | 8     | 18    | 15    | 17    | 13    | 58    | 72    | 21   |
|                                                                                  | PrecStd     | 199    | 65    | 61    | 52    | 43    | 47    | 35    | 44    | 57    | 63    | 65    | 67    | 68   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.9  | 13.2  | 16.8  | 17.9  | 22.0  | 23.4  | 23.3  | 23.1  | 21.1  | 17.9  | 15.4  | 13.9 |
|                                                                                  |             | MCWB   | 11.8  | 11.2  | 10.6  | 12.3  | 15.7  | 17.1  | 18.0  | 18.2  | 16.0  | 14.9  | 13.6  | 12.6 |
|                                                                                  | 2%          | DB     | 12.1  | 12.1  | 13.1  | 15.3  | 18.3  | 20.3  | 20.4  | 19.9  | 19.0  | 16.6  | 14.3  | 13.1 |
|                                                                                  |             | MCWB   | 11.2  | 10.9  | 10.0  | 11.0  | 13.8  | 15.4  | 16.8  | 17.0  | 15.6  | 14.5  | 13.1  | 12.0 |
|                                                                                  | 5%          | DB     | 11.6  | 11.5  | 12.0  | 13.5  | 16.2  | 18.3  | 18.8  | 18.6  | 17.9  | 15.9  | 13.5  | 12.4 |
|                                                                                  |             | MCWB   | 10.6  | 10.4  | 10.2  | 10.6  | 12.8  | 14.8  | 16.1  | 16.1  | 15.4  | 14.2  | 12.3  | 11.4 |
|                                                                                  | 10%         | DB     | 11.0  | 10.9  | 11.4  | 12.6  | 14.7  | 16.8  | 17.8  | 17.7  | 17.1  | 15.2  | 12.9  | 11.7 |
|                                                                                  |             | MCWB   | 10.1  | 9.8   | 9.8   | 10.3  | 12.0  | 14.1  | 15.6  | 15.6  | 15.1  | 13.7  | 11.7  | 10.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.2  | 11.9  | 11.9  | 13.2  | 16.1  | 18.1  | 18.8  | 19.2  | 17.7  | 16.0  | 14.2  | 13.1 |
|                                                                                  |             | MCDB   | 12.5  | 12.2  | 12.7  | 15.8  | 20.4  | 22.6  | 22.1  | 22.0  | 19.0  | 16.9  | 14.7  | 13.5 |
|                                                                                  | 2%          | WB     | 11.6  | 11.2  | 11.3  | 12.1  | 14.5  | 16.2  | 17.5  | 17.7  | 16.9  | 15.3  | 13.5  | 12.3 |
|                                                                                  |             | MCDB   | 11.9  | 11.7  | 12.2  | 13.6  | 17.2  | 18.7  | 19.3  | 19.1  | 17.8  | 16.1  | 14.1  | 12.8 |
|                                                                                  | 5%          | WB     | 11.0  | 10.7  | 10.8  | 11.4  | 13.4  | 15.3  | 16.7  | 16.9  | 16.2  | 14.7  | 12.8  | 11.7 |
|                                                                                  |             | MCDB   | 11.4  | 11.3  | 11.7  | 12.8  | 15.3  | 17.2  | 18.1  | 17.9  | 17.2  | 15.5  | 13.3  | 12.2 |
|                                                                                  | 10%         | WB     | 10.4  | 10.0  | 10.2  | 10.8  | 12.6  | 14.6  | 16.1  | 16.2  | 15.5  | 14.1  | 12.0  | 11.0 |
|                                                                                  |             | MCDB   | 11.0  | 10.8  | 11.3  | 12.2  | 14.1  | 16.3  | 17.3  | 17.3  | 16.7  | 15.0  | 12.7  | 11.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.1   | 4.4   | 5.0   | 5.3   | 5.2   | 5.1   | 4.4   | 4.6   | 4.9   | 4.5   | 4.2   | 4.1  |
|                                                                                  |             | MCDBR  | 4.0   | 4.0   | 5.3   | 6.7   | 8.1   | 8.5   | 6.4   | 6.0   | 5.9   | 4.6   | 4.0   | 3.9  |
|                                                                                  |             | MCWBR  | 4.4   | 3.9   | 4.1   | 4.5   | 5.0   | 5.1   | 4.1   | 4.1   | 4.1   | 3.9   | 4.0   | 4.3  |
|                                                                                  | 5% WB       | MCDBR  | 3.9   | 3.8   | 4.6   | 5.4   | 6.6   | 6.5   | 5.4   | 5.1   | 4.5   | 4.0   | 3.7   | 3.8  |
|                                                                                  |             | MCWBR  | 4.5   | 4.2   | 4.3   | 4.3   | 4.5   | 4.4   | 4.0   | 4.0   | 4.1   | 4.0   | 4.1   | 4.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.361  | 0.364 | 0.384 | 0.388 | 0.400 | 0.378 | 0.384 | 0.369 | 0.368 | 0.359 | 0.351 | 0.354 |      |
|                                                                                  | taud        | 2.206  | 2.282 | 2.270 | 2.306 | 2.320 | 2.421 | 2.412 | 2.467 | 2.455 | 2.399 | 2.275 | 2.181 |      |
|                                                                                  | Ebn at Noon | 602    | 737   | 800   | 847   | 854   | 876   | 863   | 860   | 817   | 739   | 614   | 533   |      |
|                                                                                  | Edh at Noon | 73     | 88    | 107   | 116   | 120   | 110   | 109   | 98    | 89    | 78    | 67    | 63    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.74   | 1.44  | 2.63  | 4.27  | 5.16  | 5.58  | 5.03  | 4.31  | 3.11  | 1.80  | 0.91  | 0.56  |      |
|                                                                                  | RadStd      | 0.07   | 0.11  | 0.23  | 0.28  | 0.28  | 0.48  | 0.42  | 0.29  | 0.25  | 0.15  | 0.07  | 0.06  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (7 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page